

Ayoub Foussoul

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ACADEMIC APPOINTMENTS

University of Chicago Booth School of Business

Postdoctoral Researcher

2025 –

Chicago, IL

EDUCATION

Columbia University

PhD in Operations Research

2020 – 2025

New York, NY

- Advisor: Vineet Goyal

- Thesis: Robust and Tractable Policies for Resource Allocation Under Uncertainty

Master of Science in Operations Research

École Polytechnique

Master of Science in Applied Mathematics

2017 – 2020

Paris, France

Bachelor of Science in Mathematics and Computer Science

Lycée Moulay Youssef

Classes préparatoires, Mathematics, Physics, Computer Science

2015 – 2017

Rabat, Morocco

PAPERS

- [1] Discrete Input Landscapes of Neural Networks ([link](#))
Ozan Candogan & Ayoub Foussoul
Submitted to **ICLR 2027**
- [2] Neural Scaling Laws for Customer Choice Prediction
Ozan Candogan, Ayoub Foussoul & Feiyu Han
Working paper
Preliminary version accepted at the **NeurIPS 2026 MLxOR Workshop**
- [3] Deep Flow Networks: Optimization-Friendly Surrogates for Integer Predict-then-Optimize ([link](#))
Ozan Candogan & Ayoub Foussoul
Submitted to **Management Science**
Preliminary version appeared in proceedings of **ICML 2026 (spotlight paper)**
- [4] Distributionally Robust Newsvendor on a Metric ([link](#))
Ayoub Foussoul & Vineet Goyal
Major revision in **Operations Research**
Extended abstract appeared in proceedings of **EC 2025**
- [5] LP-based Approximations for Disjoint Bilinear and Two-Stage Adjustable Robust Optimization ([link](#))
Omar El Housni, Ayoub Foussoul & Vineet Goyal
Mathematical Programming
Preliminary version appeared in proceedings of **IPCO 2022**
- [6] Two-Stage Stochastic Stable Matching ([link](#))
Ayoub Foussoul, Yuri Faenza & Chengyue He
In proceedings of **IPCO 2024**

- [7] Minimum Cut Representability of Stable Matching Problems ([link](#))
Ayoub Foussoul, Yuri Faenza & Chengyue He
Major revision in *Operations Research*
★ Honorable Mention, INFORMS Optimization Society, 2025 Student Paper Prize
- [8] Fully-Dynamic Load Balancing ([link](#))
Ayoub Foussoul, Vineet Goyal & Amit Kumar
Mathematical Programming
*Preliminary version appeared in proceedings of **IPCO 2024***
- [9] Last Switch Dependent Bandits with Monotone Payoff Functions ([link](#))
Ayoub Foussoul, Vineet Goyal, Orestis Papadigenopoulos & Assaf Zeevi
*In proceedings of **ICML 2023***
- [10] MNL-Bandit in Non-Stationary Environments ([link](#))
Ayoub Foussoul, Vineet Goyal & Varun Gupta

TECHNICAL REPORTS

- [1] Conditioning of Piecewise Linear Approximations with an Application to the Monitoring of the COVID-19 Epidemic
Marianne Akian, Jérôme Bolt, Ayoub Foussoul, Stephane Gaubert & Laurent Massoulié
★ École Polytechnique Research Prize in Applied Mathematics
- [2] Identification of the Optimal Tropical Hypersurfaces Explaining Consumers' Responses to Prices
Xavier Allamigeon, Ayoub Foussoul & Stephane Gaubert
★ École Polytechnique Research Prize in Applied Mathematics

TEACHING EXPERIENCE

Optimization I, IEOR E6613 <i>Core PhD course on linear and convex optimization</i> <i>Teaching Assistant & Guest Lecturer</i>	Fall 2021, 2022
Optimization II, IEOR E6614 <i>Core PhD course on combinatorial optimization</i> <i>Teaching Assistant</i>	Spring 2023, 2024, 2025
Convex Optimization, EEOR E6616 <i>PhD course on convex optimization</i> <i>Teaching Assistant</i>	Spring 2022
Optimization Methods, IEOR E4004 <i>Core master's course on optimization</i> <i>Teaching Assistant & Guest Lecturer</i>	Fall 2023
Applications for Financial Engineering, IEOR E4500 <i>Master's course on quantitative methods in financial engineering</i> <i>Teaching Assistant</i>	Spring 2021
Academic Tutor, Le Valdocco <i>Provided academic support and mentoring to underprivileged students at a non-profit educational organization.</i>	2017 – 2018

INDUSTRY EXPERIENCE

INRIA

Mar 2020 – Sep 2020

Research Intern

Paris, France

- **Constrained piecewise-linear approximation for epidemic monitoring:** Developed a dynamic-programming method for constrained piecewise-linear approximation in L_p norms ($p \geq 1$) and applied it to French COVID-19 observables (SIVIC/SAMU) (see acknowledgments in paper).
- **VC dimension of tropical polynomials:** Studied tropical polynomial classifiers and proved that the VC dimension of tropical polynomials matches that of classical real polynomials.

Padoa

Jun 2019 – Sep 2019

Data Analyst Intern

Paris, France

Occupational-medicine platform connecting clinicians, employers, and employees

- Implemented a Health Statistics page in the clinician-facing web app.
- Extended the backend Stats API to expose the KPI/statistics required by the new dashboard.

SERVICE AND OUTREACH

Elected Student Representative, École Polytechnique (Class of 2020)

Co-organizer of IEOR PhD Tutorial Series

Reviewer: Management Science, Mathematical Programming, Optimization Letters,
NeurIPS MLxOR Workshop, INFORMS Journal on Computing, IPCO, SODA, STOC, EC, ICML

AWARDS

Honorable Mention in the INFORMS Optimization Society Student Paper Prize	2025
Postgraduate Excellence Scholarship (OCP Foundation)	2020
École Polytechnique Research Prize in Applied Mathematics	2020
French Government Major-Excellence Scholarship	2017

TALKS

Deep Flow Networks: Optimization-Friendly Surrogates for Integer Predict-then-Optimize

- INFORMS Annual Meeting, November 2026, San Francisco, CA
- LAMP Workshop at TTIC, August 2026, Chicago, IL (poster session)
- Machine Learning for Algorithms Workshop at STOC 2026, June 2026, Salt Lake City, UT (poster session)
- Chicago Operations Day, June 2026, Chicago, IL (poster session)

Distributionally Robust Newsvendor on a Metric

- INFORMS Annual Meeting, October 2025, Atlanta, GA
- Revenue Management and Pricing Conference (RMP), July 2025, New York, NY
- ACM Conference on Economics and Computation (EC), July 2025, Stanford, CA
- Manufacturing & Service Operations Management Conference (MSOM), June 2025, London
- Columbia University Data Science Day, April 2025, New York, NY (poster session)
- IEOR Colloquium, November 2024, New York, NY
- INFORMS Annual Meeting, October 2024, Seattle, WA
- Cornell ORIE Young Researchers Workshop, October 2024, Ithaca, NY

- Northwestern Kellogg Operations Management Rookiepalooza, October 2024, Evanston, IL

Two-Stage Stochastic Stable Matching

- Integer Programming and Combinatorial Optimization Conference (IPCO), July 2024, Wrocław
- Columbia University Data Science Day, April 2024, New York, NY (poster session)

Fully-Dynamic Load Balancing

- Integer Programming and Combinatorial Optimization Conference (IPCO), July 2024, Wrocław
- Columbia IEOR Student Seminar, February 2024, New York, NY

Last Switch Dependent Bandits with Monotone Payoff Functions

- International Conference on Machine Learning (ICML), July 2023, Honolulu, HI (poster session)

MNL-Bandit in Non-Stationary Environments

- DSI Financial and Business Analytics Poster Session, November 2023, New York, NY (poster session)
- INFORMS Annual Meeting, October 2023, Phoenix, AZ
- Revenue Management and Pricing Conference (RMP), July 2023, London

LP-based Approximations for Disjoint Bilinear and Two-Stage Adjustable Robust Optimization

- International Symposium on Mathematical Programming (ISMP), July 2024, Montréal
- International Research and Innovation Seminar (IRIS), December 2023, Ben Guerir
- INFORMS Annual Meeting, October 2022, Indianapolis, IN
- International Conference on Continuous Optimization (ICCOPT), July 2022, Bethlehem, PA
- Integer Programming and Combinatorial Optimization Conference (IPCO), June 2022, Eindhoven
- INFORMS Optimization Society Conference (IOS), March 2022, Greenville, SC